

Closed-loop insulin delivery—the path to physiological glucose control

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Abstract

The development of an artificial pancreas for the treatment of insulin-dependent diabetes is a highly desired endeavor for patients, physicians, scientists, and engineers. Historical algorithms and recent progress in research and technology are reviewed in the present article, together with aspects of β -cell physiology that lead to normal glucose tolerance. Algorithms are evaluated for their ability to deliver insulin as to recreate, as closely as possible, glucose and insulin profiles observed in healthy individuals. Emphasis is placed upon how the algorithms compare to the β -cell's secretory response, specifically first-phase and second-phase insulin secretion. Experimental closed-loop data employing intravenous and subcutaneous glucose sensors and implanted and external insulin pumps (Medtronic MiniMed, Northridge CA) are presented.

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Keywords: Diabetes mellitus; Insulin secretion; Insulin action; Automated insulin delivery; Artificial pancreas; Subcutaneous glucose sensor; Continuous glucose sensing; CSII; Insulin pump; Control model

Contents

1. Introduction	126
2. Approaches to closed-loop insulin delivery	126
3. Physiologic insulin delivery	127
3.1. The importance of first-phase insulin release	128
3.2. The importance of second-phase insulin release	128
3.3. Other important β -cell characteristics	129
3.4. β -Cell control compared to historical closed-loop data	129
4. Creating a physiologic insulin delivery system	132
4.1. Modeling β -cell secretion	132
4.2. Challenges specific to creating an artificial β -cell	134
4.2.1. Sensor delays	134
4.2.2. Site-specific insulin dynamics	135

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4.2.3.	Variability or lack of portal insulin effect	136
4.2.4.	Inability to effect counter-regulation	137
5.	Closed-loop experimental results	137
5.1.	Subcutaneous glucose sensing with external insulin delivery	137
5.2.	Intravenous glucose sensing with intraperitoneal insulin delivery	140
6.	Conclusions.	140
	References	141

1. Introduction

...If someone says that the task is “mechanical” it does not mean that people are incapable of doing the task; it implies though, that only a machine could do it over and over without ever complaining, or feeling bored. Douglas Hofstadter, Gödel, Escher, Bach: An Eternal Golden Braid

The treatment of insulin-dependent diabetes mellitus is both difficult and tedious, prompting G. Reach to begin his 1989 editorial comments on the artificial pancreas, aptly subtitled “...the gap between the dream and reality” [1], with the above Douglas Hofstadter passage. Yes, many individuals with type 1 diabetes can and do manage the day-to-day task of administering insulin very well. Nonetheless, the desire and demand for an automated artificial pancreas has existed almost from the time insulin was discovered in 1921. However, in addition to insulin, such a system requires a pump, a glucose sensor, and an algorithm—three simple requests.

The first commercially available insulin pumps for ambulatory use appeared on the market around 1980, approximately 60 years after insulin was first discovered. Glucose sensor technology for continuous measurement, while promised in the 1980s, is only now arriving [2–8]. In spite of the long lag time developing sensors and pumps, research into the insulin delivery algorithms has continued unabated—relying primarily on computer simulations of insulin action and controller design. Undoubtedly though, the lack of a readily available glucose sensor has limited the ability of researchers to fully investigate and validate the control strategies proposed.

As glucose-sensing technology is now catching up with insulin delivery technology, the potential to develop an ambulatory automated closed-loop system

seems finally to have arrived. With this in mind, it seems useful to review what has already been done to develop an algorithm, to reevaluate what the desired characteristics of a closed-loop system might be, and to reassess how to achieve the desired goals.

2. Approaches to closed-loop insulin delivery

Many control algorithms have already been proposed for regulation of blood glucose by insulin. These include an early on–off algorithm by Kadish [9], the Biostator algorithms [10–12], and algorithms proposed by Fischer [13,14], Shichiri [8,15], Kraegen [16], and Albisser [17]. Later algorithms, using more advanced control theory, include a one-step-ahead adaptive controller by Radziuk [18], a model predictive controller by Parker [19], a neural network controller by Trajanoski [20], and an H-infinity controller by Kienitz [21]—many of which have been recently reviewed by Parker et al. [22,23]. Rather than rereview all the control strategies here, we focus on aspects of the design process that are common to most of the algorithms. Essentially—and this is true for control theory in general—design begins with a model of the “system being controlled” and proceeds to optimize the control based on prespecified “desirable characteristics.” There are two main issues with this approach as it pertains to closed-loop insulin delivery: validation of the model being controlled (glucose-metabolism/insulin-action), and specifying the desirable characteristics (what is an optimal glucose excursion?).

Models describing insulin action in vivo include the minimal model developed by Bergman et al. [24–26], a modified minimal model shown to be more accurate in determining hepatic glucose output [27–29], a model developed in Karlsburg [14], and a model developed by Sorenson et al. [30]. However,

as the models differ in structure—in some cases quite dramatically—the differences can be expected to affect conclusions regarding the control strategy. Thus, at a minimum, if a model of insulin action/glucose appearance is used to develop the control strategy, the underlying assumptions and predictions of the model as it pertains to the closed-loop may need further validation. Ideally, differences in the various model predictions would be reconciled with experimental data.

In addition to model differences, each of the proposed methods requires optimization criteria to be specified. For example, one-step-ahead adaptive control always requires the next value of glucose to be specified, and model predictive control requires all future values leading to euglycemia to be specified. If minimizing the magnitude of the glucose excursion is the only criterion, many algorithms can be devised that will perform “better” than the β -cell [31] (see

Fig. 5 for an example). However, as insulin has both metabolic and mitogenic effects, delivery algorithms that are designed to minimize the glucose excursions without regard for how much insulin is delivered may cause excessive weight gain, hypertension, or arteriosclerosis [32–34].

3. Physiologic insulin delivery

An alternative to using a model of insulin action and specifying an optimized glucose excursion is to understand how the β -cell controls glucose and to subsequently use this information to determine how an artificial closed-loop algorithm should behave. In vivo, the β -cell responds to glucose challenges with a characteristic “first” and “second” phase response [35,36]. The biphasic response is clearly seen during hyperglycemic clamps such as shown in Fig. 1a (data

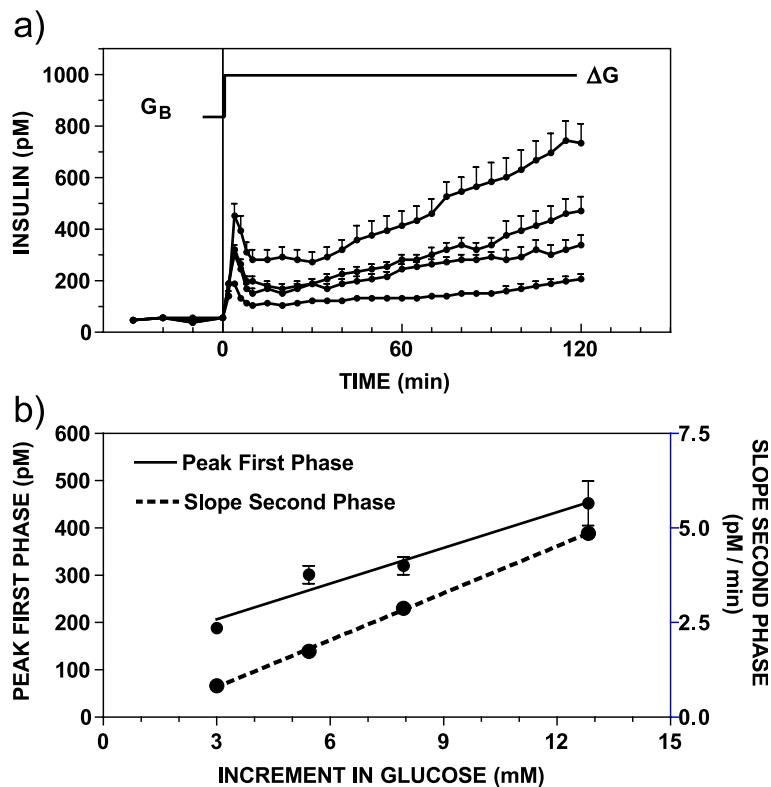


Fig. 1. (a) Plasma insulin concentrations during hyperglycemic clamps performed with increasing glucose levels (ΔG). (b) Peak first-phase response (left axis; solid line) and slope of second-phase (dashed line) versus ΔG (data adapted and reprinted with permission from *The American Diabetes Association* from Ref. [35]).

adapted from Elahi et al. [35]). In addition, the underlying linearity of the β -cell response can be determined from this data. The peak first-phase insulin concentration and the rate of increase of second-phase insulin (slope) are both proportional to the increment in plasma glucose (Fig. 1b).

3.1. The importance of first-phase insulin release

The importance of first-phase insulin release has been shown in many studies (reviewed in Refs. [37,38]) and defects in the response have been linked with the etiology of type 2 diabetes (reviewed in Ref. [39]). Improvements in first-phase response have been linked to improvements in glucose tolerance [40,41]. Generally, an early insulin response results in less total insulin needed to yield a similar area under the glucose excursion curve since glucose uptake is proportional to glucose concentration times insulin effect [42]. First-phase insulin is thought to result in an early suppression of hepatic glucose output [38]. The importance of first-phase insulin has been emphasized by studies indicating that, in subjects with normal glucose tolerance, the product of first-phase insulin release (ϕ_1) and insulin sensitivity (S_I) is constant, known as the disposition index, $DI = \phi_1 S_I$ [43]. This adaptation is shown in Fig. 2 for a subject who remained normal glucose tolerant (closed circles) while slowly becoming insulin resistant (solid line indicates the fit for $DI = \phi_1 S_I$; data adapted from Refs.

[44,45]). A subject who became insulin resistant with no adaptation in insulin secretion—i.e., became diabetic—is shown for comparison (triangles). The tight relation between insulin sensitivity and insulin release in the subject who did not become diabetic clearly indicates the β -cell can adapt its “control gain” to maintain a desired level of glucose tolerance.

3.2. The importance of second-phase insulin release

Although the importance of the first-phase insulin response has been more heavily emphasized in the literature, the contribution of second-phase cannot be overlooked. From a closed-loop control perspective, the continued increase in plasma insulin during constant hyperglycemia should establish the necessary basal insulin delivery rate to maintain fasting plasma glucose at setpoint irrespective of insulin sensitivity or rate of endogenous glucose appearance. Controllers that exhibit this property are referred to as “type 1” systems [46]. Both first- and second-phase insulin release have been shown to adapt to insulin sensitivity, as can be seen in the insulin responses of highly exercise-trained individuals compared to normal controls (Fig. 3, adapted from Ref. [36]). In the exercise-trained individuals, the insulin response was approximately half that of normal controls, while at the same time, glucose uptake was matched. This suggests that the two groups had identical disposition indices, i.e., the exercise-trained individuals had twice the insulin

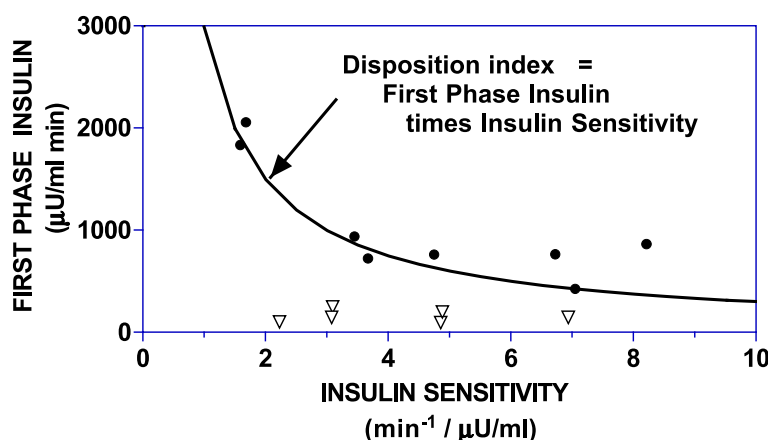


Fig. 2. Adaptation of first-phase insulin release (ϕ_1) to a progressive loss in insulin sensitivity (S_I) over a 25 year period in a subject who did not develop diabetes (closed circles; data adapted from Refs. [44,45]). Solid line represents the best fit to the hyperbolic relation $DI = \phi_1 S_I$. A subject who became insulin resistant with no adaptation in insulin secretion is shown for comparison (triangles).

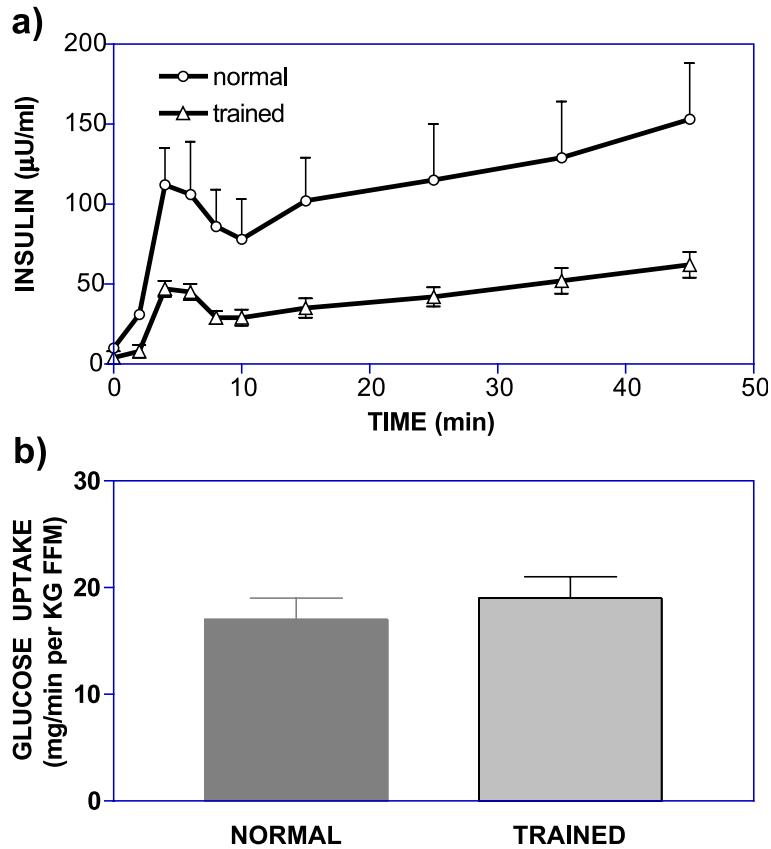


Fig. 3. (a) Insulin secretion during hyperglycemic clamps in exercised-trained individuals (circles) and healthy controls (triangles). (b) Glucose uptake during the same clamps (data adapted with permission from Ref. [36]).

sensitivity and 1/2 the insulin response, leading to identical glucose uptake rates. Again, this suggests that the β -cell seeks an “optimal” glucose tolerance level.

3.3. Other important β -cell characteristics

Although the first-phase/second-phase characterization captures two important aspects of the β -cell response, it is clearly an incomplete description. Other known characteristics of the β -cell include its ability to adjust secretion based on antecedent hyperglycemia, the so-called “glucose priming” effect [47], and its tendency to oscillate or release insulin in discrete pulses [48]. Further, the β -cell sensitivity to glucose may be altered depending on free-fatty-acid (FFA) levels [49]—a point that may manifest itself in higher secretory gains at breakfast relative to lunch or dinner. Prior to meals, the β -cell is stimulated to secrete insulin by

neural signals [50,51], and during the meal, it is stimulated by gut hormones [52], and finally, β -cell secretion may be inhibited by the prevailing concentration of insulin in plasma [53–57]. The relative importance of each of these characteristics in maintaining physiologic glucose control may need to be considered in the context of an artificial β -cell system.

3.4. β -Cell control compared to historical closed-loop data

The most widely known of all insulin delivery algorithms, for which experimental data are available, are those attributed to the Biostator [10–12] (detailed in Fig. 4 along with algorithms developed by Albisser [58], Kraegen [16,59], and Fischer [13]). The performance of these algorithms has previously been compared in two studies [17,60]. Here, we focus on

1. Biostator 1

$$ID = R \cdot \left(\frac{G^n + DF - GB}{Q} + 1 \right)^4$$

$$DR = \frac{KR \cdot A^2}{10} + 6 \cdot A \quad \text{if } (A > 0)$$

$$DF = -\frac{KF \cdot A^2}{10} + 6A \quad \text{if } (A \leq 0)$$

$$A = \frac{4 \cdot G^n - G^{n-1} - G^{n-2} - G^{n-3}}{10}$$

2. Biostator 2

$$ID = R \left(\frac{A - GB}{Q} + 1 \right)^2 \begin{cases} 1 & \text{if } (A - GB) > 0 \\ 0 & \text{if } (A - GB) \leq 0 \end{cases}$$

$$+ R \cdot m \cdot (A - GB) \begin{cases} KR & \text{if } (A - GB) > 0 \\ KF & \text{if } (A - GB) \leq 0 \end{cases}$$

$$m = \frac{2 \cdot G^n + 1 \cdot G^{n-1} - 1 \cdot G^{n-3} - 2 \cdot G^{n-4}}{10}$$

$$A = 2 \cdot m + \frac{G^n + G^{n-1} + G^{n-2} + G^{n-3} + G^{n-4}}{5}$$

3. Kraegen

$$ID = \frac{1}{2} M [1 + \tanh S(G^n + DF + B)]$$

$$DF = K_1 e^{(A/K_2 - 1)}$$

$$A = \frac{8 \cdot G^n - 4 \cdot G^{n-1} - 2 \cdot G^{n-2} - 1 \cdot G^{n-3}}{15}$$

4. Albisser

$$ID = \frac{1}{2} M [1 + \tanh S(G^n + DF + B)]$$

$$DF = K_1 A^2 + K_2 A$$

$$A = \frac{4 \cdot G^n - G^{n-1} - G^{n-2} - G^{n-3}}{10}$$

5. Fischer

$$ID = k_0 + k_1 (G^n - G_b) + k_2 A$$

$$A = G^n - G^{n-1}$$

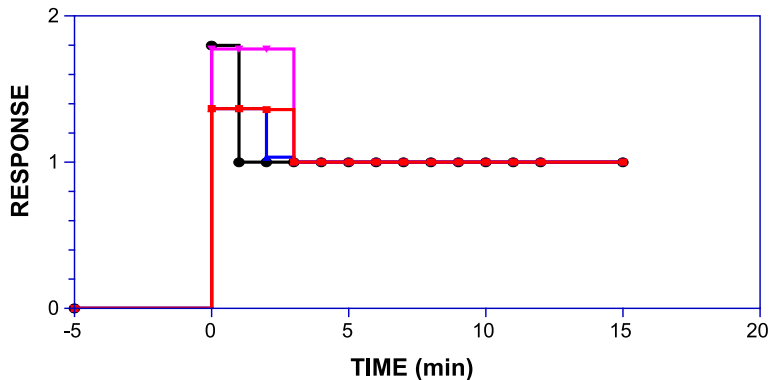


Fig. 4. Classical control algorithms proposed for closed-loop insulin delivery (left column). Finite difference formulations used for estimating the derivative response (right column; G^n , G^{n-1} refers to the most recent glucose value, the glucose value 1 min back in time, etc.). Simulated step responses of each algorithm (bottom panel).

aspects of the algorithms that can be concluded based on examination of the algorithm equations per se.

First, the step response—which is easily simulated (Fig. 4, bottom)—can be compared to the β -cell's response to a hyperglycemic clamp. From this rather simple analysis, it is evident that each algorithm has a “ β -cell like” first phase but lacks the rise in “second phase” (compare to Fig. 1). A second phase exists but remains constant.

Second, in each of the formulations, the duration of the first-phase response is altered, spanning 2–5 min. This duration is likely consistent with the β -cell and is determined by the terms in the right-hand column of Fig. 4 (terms involving A and m). Note that the terms used to obtain the derivative all have the same structure (weighted history of past glucose

values). This structure is known in digital signal processing as a finite impulse response filter (FIR filter) where the number of terms back in time is the filter order. The order and weighting coefficients determines the filter's response time and sensitivity to noise. The filters of Fig. 4 can all be shown to be highly sensitivity to noise. Increasing the filter order would have reduced the sensitivity to noise but at the expense of delaying the response. The sensitivity to noise implies that all the algorithms would have reacted to artifactual increases in glucose with insulin delivery. This sensitivity may have limited the relative amount of insulin delivered by this mechanism (limited the ratio of first to second phase). Although two of the algorithms (Biostator and Fischer) allowed the user to adjust this ratio, the published ranges were

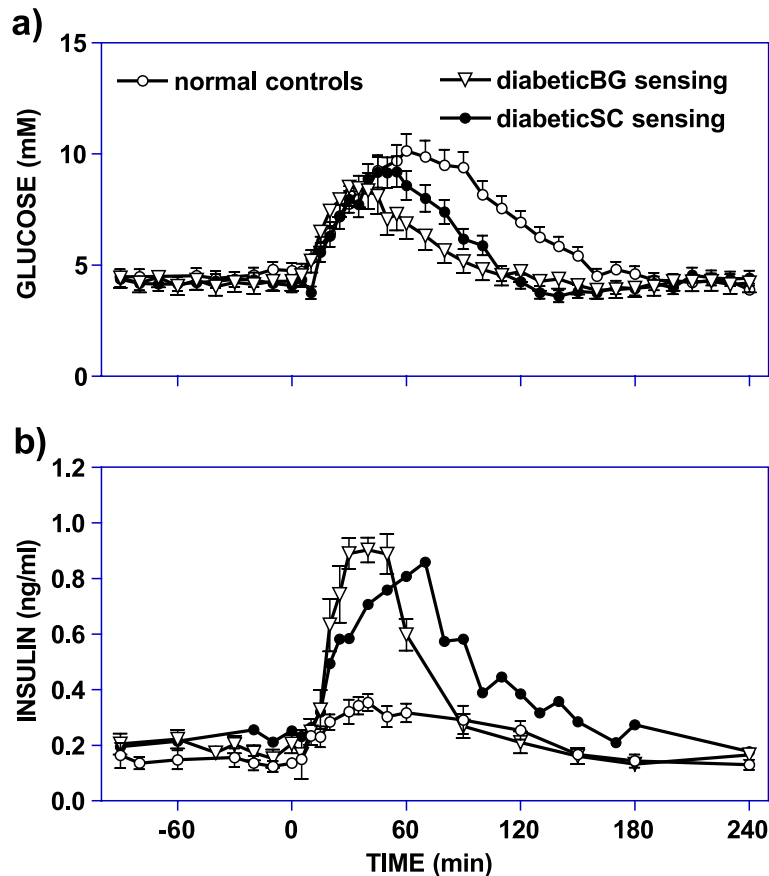


Fig. 5. Plasma glucose (a) and insulin (b) levels in normal dogs (open circles) and diabetic dogs under PD control with a sensor exposed to interstitial fluid (closed circles) or blood (triangle; biostator-like device). Dynamics were obtained during oral glucose tolerance tests (data adapted with permission from Ref. [31]).

only from ~ 2 to 11 min for all algorithms. These ratios are likely significantly less than the β -cell is believed to use [61].

The third and final observation from this data is, with the exception of the Fischer proportional-derivative (PD) algorithm, that all the algorithms have nonlinear dose responses. For example, if the increment in glucose were doubled, the original Biostator algorithm would increase second phase by a factor of 16 (the response goes up as a power of 4) and first phase by a factor of approximately 4 (proportional to $A^2 + A$). Although these relations are not necessarily bad, they do disagree with the β -cell (Fig. 1b).

Of the algorithms in Fig. 4, the PD algorithm [17] has been most rigorously compared to the β -cell. Glucose and insulin responses during OGTTs performed in normal dogs were compared to the responses obtained in diabetic dogs under PD control (Fig. 5; data adapted from Ref. [31]). For the experiments performed in the diabetic dogs, a sensor was either implanted in the SC tissue or directly exposed to blood (biostator-like circuit). Two observations can be made from this data. One, irrespective of whether blood glucose or sensor glucose was used to drive the algorithm, the insulin response was considerably higher than the β -cell's response, and the glucose response was considerably lower than that observed in normal controls. This suggests that if lower gains had been used (a_1 and a_2 —equivalent to K_P and K_D in PD control notation), a more physiologic response would have been obtained. Second, when the SC sensing site was used, both the plasma glucose and insulin responses were slightly higher than when the same sensor was exposed to blood. This emphasizes the need to minimize or compensate for delays in the feedback loop.

4. Creating a physiologic insulin delivery system

Medtronic MiniMed is developing external and implantable insulin infusion pumps, as well as external (SC) and implantable (IV) glucose sensors. The long-term goal is to offer both external and fully implanted closed-loop insulin delivery systems. The algorithms being developed rely on understanding and emulating how the β -cell establishes its glucose control. Digital signal processing and control systems theory are used

to address specific issues related to using pumps that do not deliver insulin into the portal vein and sensors that do not directly measure blood glucose. While a completely automated system is not yet available, this section reviews work completed to date.

4.1. Modeling β -cell secretion

The β -cells' biphasic insulin response (Figs. 1 and 3) closely parallels what is probably the most widely used, and oldest, of all control algorithms. The algorithm is based on a classical proportional, integral, and derivative scheme (PID; in the present review PID also refers to Physiologic Insulin Delivery). The proportional-derivative controller proposed by Fischer (cf. Fig. 4) [13] is a subset of this class of controllers, as is a recently published proportional-integral algorithm using the Medtronic MiniMed SC sensor in a critical care application [62]. The algorithm is known to be applicable to a wide variety of dynamic systems (automobiles, planes, liquid level systems, satellite tracking devices, etc.). Insulin delivery ($ID(t)$) predicted by PID-controller is described by:

$$PID(t) = K_P(G - G_B) + K_I \int (G - G_B)dt + K_D \frac{dG}{dt} \quad (1)$$

and diagrammed in Fig. 6 for a hyperglycemic clamp. Parameters K_P , K_I , and K_D indicate the relative weights of the proportional, integral, and derivative components, and G and G_b represent blood glucose and basal glucose. Of the individual components, the derivative response produces the well-known first-phase insulin release (Fig. 6a and d) and the integral response produces the steady rise (Fig. 6a and c). As shown, the responses include low-pass and derivative filters, resulting in some delay in the proportional and integral responses and spreading of the derivative response.

From a control-systems perspective, the PID algorithm is generally considered to be quite primitive or simplistic, whereas algorithms based on optimal or model-predictive strategies are considered state of the art or advanced. Nonetheless, the PID algorithm remains the first choice of controller design in many practical applications, and it is interesting that nature appears to have emulated it in its design of the β -cell.

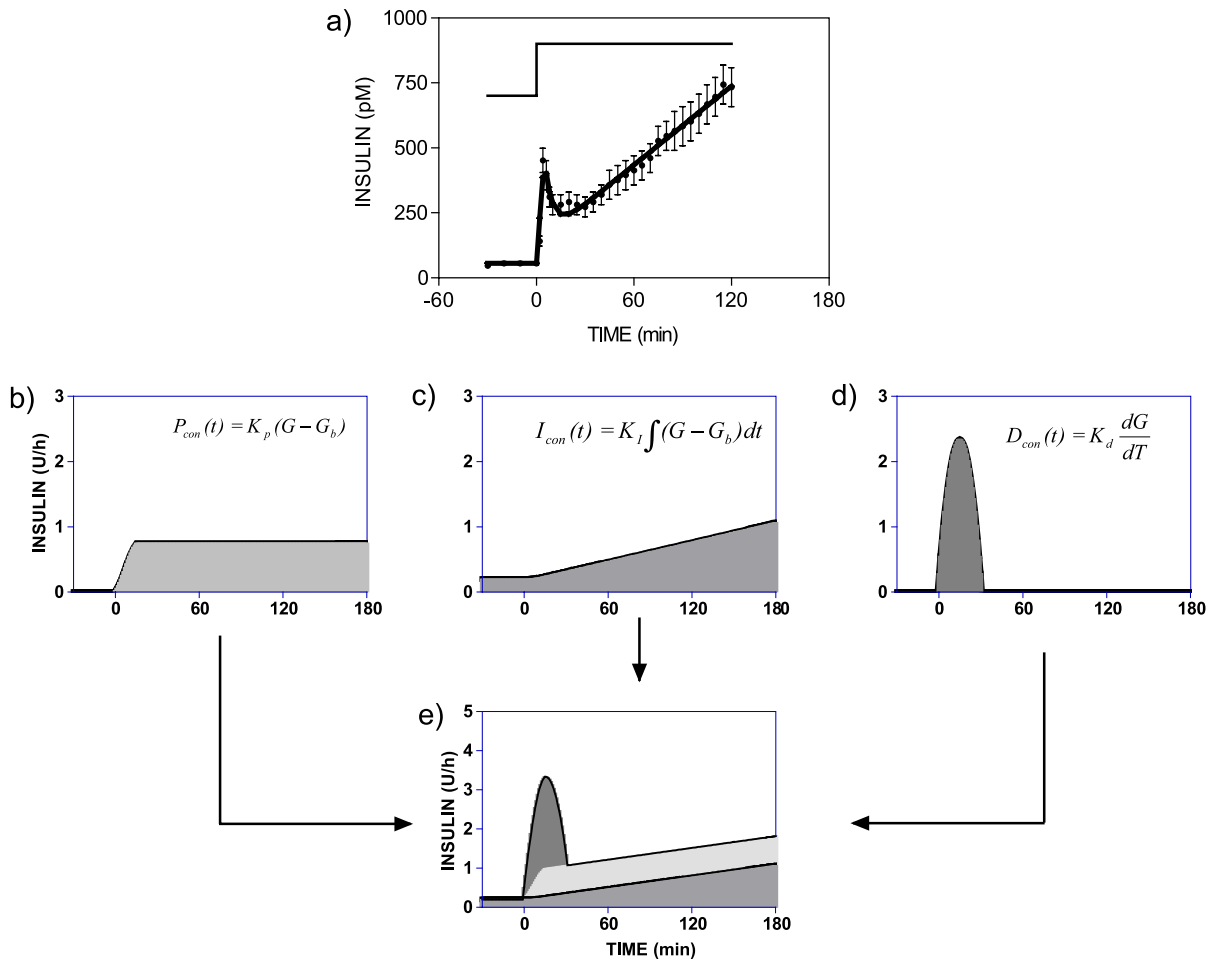


Fig. 6. (a) Response of the β -cell (closed circles) to a step change in glucose (inset) together with PID model response (Eq. (1); solid line) assuming a one-compartment model of insulin clearance. Proportional (b) integral (c) and derivative (d) components making up the total PID response (e). The one-compartment model of insulin clearance was assumed to have a fraction clearance rate of 7 min (time constant), and the individual responses (b–d) were obtained with low-order FIR filters.

From a modeling perspective, the PID model is similar to several existing secretion models [63–71], but with several key differences. Of the existing models, the one developed by Cobelli et al. [64–69] is best characterized. The Cobelli model separates the β -cell response into static and dynamic components where the dynamic component reacts to the rate of increase in plasma glucose and is zero when the glucose level is stable or falling, and the static component provides a delayed response to a steady-state increase in plasma glucose. The PID model differs from the Cobelli model in that PID secretion is suppressed during rapid falls in glucose, and there

is both an immediate proportional response to hyperglycemia and a delayed component. Both the Cobelli model and the PID model characterize the β -cell's rapid rise in response to a hyperglycemic clamp; however, under closed-loop conditions simulated by the minimal model, only the PID model adapts to changes in insulin sensitivity or rates of endogenous appearance without steady-state errors in fasting glucose (unpublished data). The PID model is also more stable due to its inclusion of a proportional response and the suppressive effect of the derivative on falling glucose values (unpublished data).

4.2. Challenges specific to creating an artificial β -cell

The ability to recreate the β -cell step response with a secretion model does not, in itself and of itself, ensure that adequate control will be achieved during day-to-day use. The model must ultimately be able to recreate the insulin-secretory profile for meal and exercise profiles and be able to use a glucose sensor as input. Depending on the type of sensor and its location, different delays and noise will be present. Similarly, depending on the type of pump and its location (subcutaneous or intraperitoneal), the insulin dynamics will be different. Finally, a functioning β -cell can be sure of a robust counter-regulatory response should it over-deliver insulin. This may not be true of individuals with type 1 diabetes where the counter-regulatory system is often compromised [72].

4.2.1. Sensor delays

All glucose-sensing technologies introduce some delay and noise into the glucose measurement (for recent reviews, see Refs. [3–5,73]). Characterizing these delays is critical for the design of a sensor-driven glucose-control system, as delays are well known to destabilize feedback loops. While sensor delay is often reported, its definition is unclear [4]. Some groups use the time between peak plasma- and sensor-glucose [74], and others define the delay as the time shift that generates the best fit between blood glucose and sensor glucose [75]. This can be done by cross-correlation or shifting the time axis until the mean square error or mean absolute difference is minimized. In the context of control systems theory, delay is traditionally separated into “time constants” and “transport lags” [46]. Time constant (τ) defines an immediate proportional response (one time constant achieves 63% equilibration; three time constants reaches 95%) and transport lag (T_D) defines an interval prior to any response being observed. For a sensor signal ($S(t)$) reacting to glucose ($G(t)$) with sensitivity α (nA per mg/dl), this would be described as

$$\tau \frac{dS(t)}{dt} = -S(t) + \alpha G(t - T_D) \quad (2)$$

Systems with time constants are generally considered easier to control than those with a transport lag [46].

Of the two glucose sensors presently being developed by Medtronic MiniMed, the SC glucose sensor has previously been characterized in vivo as having no transport lag and a time constant of between 2 and 12 min. This time constant is thought to be primarily related to interstitial glucose equilibration [4,76,77]. The IV sensor, which might be expected to react faster as it is directly exposed to blood, is actually slower (Fig. 7) and has both a time constant and a transport lag (Fig. 7b). The longer delay is attributed to the large glucose oxidase pad necessary to ensure long sensor life (greater than 1 year).

Sensor signals obtained under day-to-day use require low-pass filters to produce smooth response curves and derivative filters to estimate the rate of change. These filters can dramatically affect control performance when used to drive insulin delivery. Fortunately, the filters can be optimized for closed-loop performance in their ability to reduce noise, estimate the rate of change, and—if needed—to correct for delay.

Reduction of noise typically involves the use of FIR (finite impulse response) or IIR (infinite impulse response) filters [78]. FIR filters are a weighted average of past signal values (cf. Fig. 4), whereas IIR filters are a weighted average of past signal values plus past filter values. Many texts and articles exist discussing the advantages and disadvantages of each strategy. Here, we merely point out that the design of these filters works well when the noise and sensor-glucose have different bandwidths. The problem is that good estimates of the glucose bandwidth are not readily available in the literature, although some information does exist on the maximal rate of change of glucose [79]. Filters currently developed by Medtronic MiniMed assume that blood glucose levels have a bandwidth of ~ 1.5 –3 cycles/h.

Derivative filters are commonly derived from the FIR structure (all of the algorithms of Fig. 4 were of this form), although some IIR forms exist. The derivative filter determines the duration of the first-phase insulin response and its sensitivity to noise. The most common design for FIR derivative filters is that proposed by Savitzky–Golay (routines for designing such filters are commonly available in commercially available software such as Matlab; MathWorks, MA). Derivative filters currently being developed by Medtronic MiniMed are based either on this design (cf.

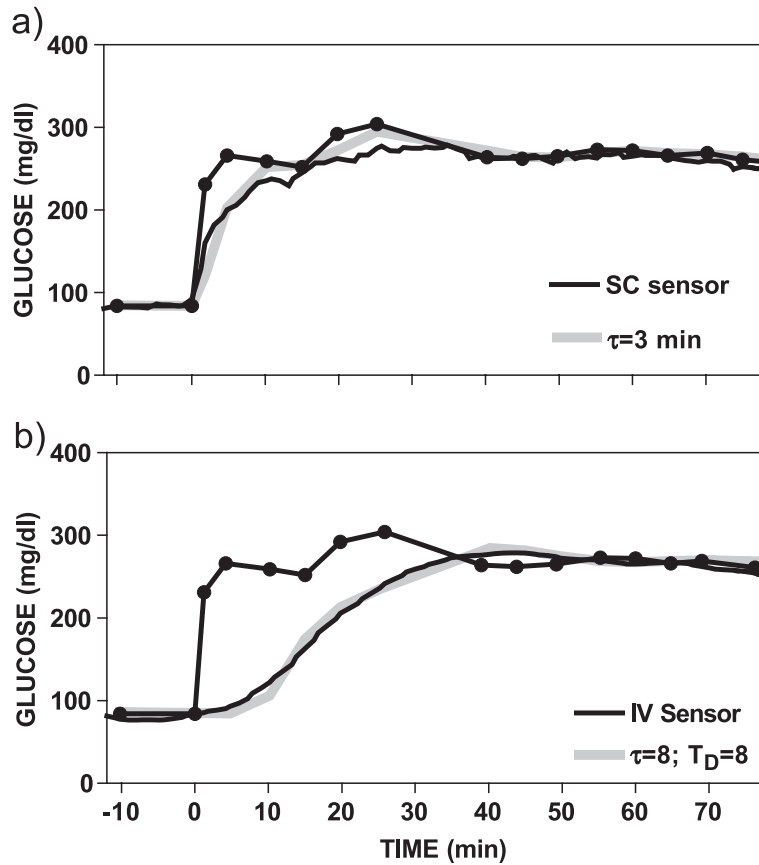


Fig. 7. (a) Response of the Medtronic MiniMed SC sensor (thin line) to a step change in plasma glucose (closed circles), (b) response of the Medtronic MiniMed IV sensor (thin line) to the same change in plasma glucose (closed circles). Thick shaded line indicates fit of Eq. (3) (τ and T_D indicate time constant and transport delay).

Fig. 4) or a least-squares frequency approach with cutoff equal to the low-pass filter cutoff.

Correction of sensor delay, which generally leads to more advanced filter designs, is accomplished with inverse or predictive filters [80]. These filters require an underlying model of sensor dynamics (Eq. (2) can suffice for this purpose). Wiener filtering [80] can be viewed in this class, as can the corrective filter strategy proposed by Schmidtke et al. [74].

4.2.2. Site-specific insulin dynamics

Insulin absorption delays will also affect closed-loop system performance. In the case of SC insulin delivery, rapid acting insulin analogues will improve the absorption rate, but challenges such as site loss [81] and reproducibility [82] remain. Intraperitoneal

insulin delivery generally results in faster absorption but is still slower than would be expected with IV delivery (Fig. 8).

There are several classical approaches to compensating for delays introduced into a feedback-loop (either sensing delays, Fig. 7, or insulin delays, Fig. 8) [46]. In regard to PID, one approach being investigated is to simply retune the ratios K_P , K_I , and K_D to best fit the plasma insulin profile obtained in normal glucose-tolerant individuals during day-to-day activity.

A second approach being investigated is to reduce the PID delivery rate by an amount proportional to the expected plasma insulin level. This latter approach is motivated by numerous studies indicating the β -cell is suppressed in proportion to plasma insulin levels [55–57]. Although the β -cell can likely sense the plasma

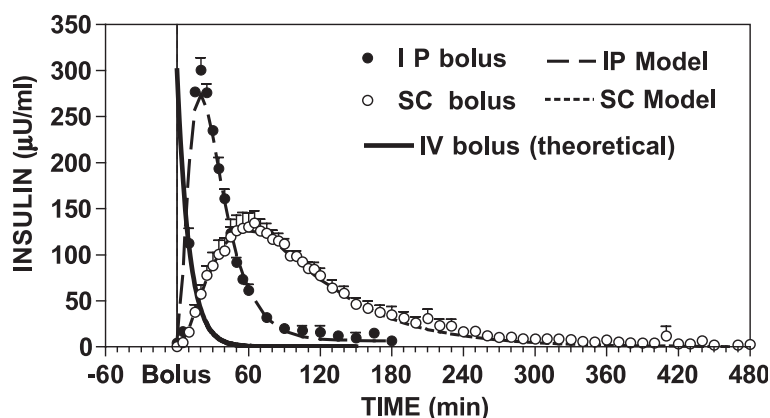


Fig. 8. Comparison of plasma insulin profiles obtained after either SC (open circles) or IP (solid circles) insulin injection in dogs. Responses from individual experiments were fit to a three-exponential model (dashed and dotted lines). A theoretical response to an IV bolus is shown for comparison (one exponential with 10-min time constant).

insulin level directly, the concentration can potentially be calculated based on a compartmental model approach (the exponential fits of Fig. 8 can be considered to be compartmental models). The predicted level ($\hat{I}_p$) could then be fed back to the controller to inhibit further insulin release according to

$$\text{PIDc}(t) = \text{PID}(t) - \gamma \hat{I}_p(t) \quad (3)$$

Here, γ is used to determine the extent to which the prediction of plasma insulin inhibits the PID output (PIDc(t) is the output with compensation). From a control systems perspective, the effect of such a feedback term is to relocate an undesirable (slow) time constant to a more desirable (fast) location. This can be verified by Laplace transform but is illustrated here by a numerical example (Fig. 9). In the example, an idealized meal response (Fig. 9a) is used as the input to the PID algorithm leading to a rapid plasma response (Fig. 9b, thick line). When the same secretion profile is applied with SC insulin delivery, the plasma insulin profile is prolonged by the absorption delay (Fig. 9b, thin line). If, however, K_p , K_I , and K_D are readjusted (least-squares fit), the ability to recreate the plasma profile achieved by the β -cell is greatly improved (Fig. 9, dotted line). The same process applied with feedback of the predicted plasma profile (Eq. (3), plasma insulin predicted assuming a two-compartment dynamics) results in still greater ability to reproduce the plasma insulin kinetics (dashed line).

4.2.3. Variability or lack of portal insulin effect

The β -cell secretes insulin directly into the portal vein from which it has a putative, direct effect to suppress hepatic glucose output (HGO) [83,84]. This led to the early conclusion that treatment of IDDM would necessarily require elevated peripheral insulin levels to restrain HGO [85]. The magnitude of a “direct portal” effect on HGO has, however, been challenged for many years [86–90] based on studies showing suppression is mediated by peripheral signals—possibly FFA [90,91]. The “direct portal” versus “indirect peripheral” controversy has largely played out using animal studies in which HGO is assessed with exogenous portal or peripheral insulin administration (portal vein catheter), and in human studies where insulin is given peripherally and tolbutamide is used to effect a β -cell response matching that obtained with peripheral insulin infusion [92,93].

The motivation behind these studies has not been to optimize insulin therapy for type 1 diabetes, but rather to understand hepatic insulin resistance and fasting hyperglycemia associated with type 2 diabetes. Few studies have assessed endogenous glucose production in type 1 individuals (assessment of hepatic glucose output/glucose turnover requires the use of glucose tracers). Studies on type 1 diabetes have primarily relied on controlling glucose in animal models with exogenous insulin delivered into the portal vein directly or into a peripheral vein, or by studies in humans

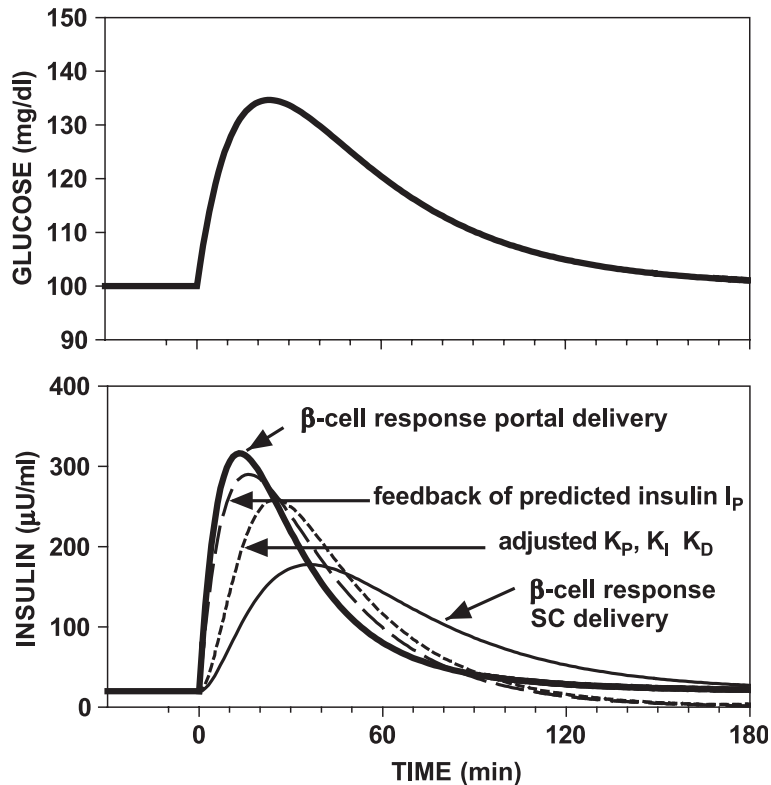


Fig. 9. Simulated meal glucose response (a) and the predicted plasma insulin profiles (b) obtained with portal insulin delivery (thick line), PID response if the insulin is delivered SC with the same gains (thin line), response obtained if K_P , K_I , and K_D are optimally readjusted (dotted line) and the response if the predicted plasma insulin profile is fed back to the PID controller (dashed line).

using an intraperitoneal versus a subcutaneous pump (cf. paper by Canavan et al. for a review [94]). At issue is whether glucose turnover can be controlled by peripheral insulin without resorting to peripheral hyperinsulinemia? Another means of addressing this issue is to compare individuals with well-controlled type 1 diabetes to normal glucose-tolerant controls. In a study conducted in pregnancy, no differences in plasma insulin, glucose, or glucose turnover were observed either during the pregnancy or postpartum [95].

4.2.4. Inability to effect counter-regulation

Neither the β -cell nor an algorithm emulating the β -cell has the ability to remove insulin once it has been delivered. However, it is likely that the β -cell can rely on a robust counter-regulatory response should it over-deliver insulin. In individuals with type 1 diabetes, this response is often compromised [72], suggesting that the artificial closed-loop system response may need to

be less aggressive. In the controls sense, “less aggressive” likely means “over-damped.” Under- versus over-damping is determined by the closed-loop control gain, or in this case, insulin sensitivity times insulin delivery (cf. the disposition index of Fig. 5). Thus, for a given level of insulin sensitivity, the algorithm gain may need to be lower than what the β -cell would choose so as to maintain safe margins of insulin delivery. The setpoint for glucose may also need to be elevated.

5. Closed-loop experimental results

5.1. Subcutaneous glucose sensing with external insulin delivery

In vivo closed-loop studies in dogs using a SC glucose sensor (Medtronic MiniMed CGMS) and an

external insulin delivery pump (Medtronic MiniMed 511 Paradigm) have been performed. In an early experiment designed to establish a “tuning hypothesis” (later validated in a dose–response study [96]), control was initiated under hyperglycemic conditions (Fig. 10). Hyperglycemia was chosen as the starting condition to validate that the integral component of

the PID algorithm could establish the necessary basal rate to achieve euglycemia (setpoint = 120 mg/dl). Meals were administered at noon of the first day—after the algorithm established setpoint—and in the morning on the second day. The ratios of first and second phase were determined by fitting the PID model to plasma insulin profiles observed in normal

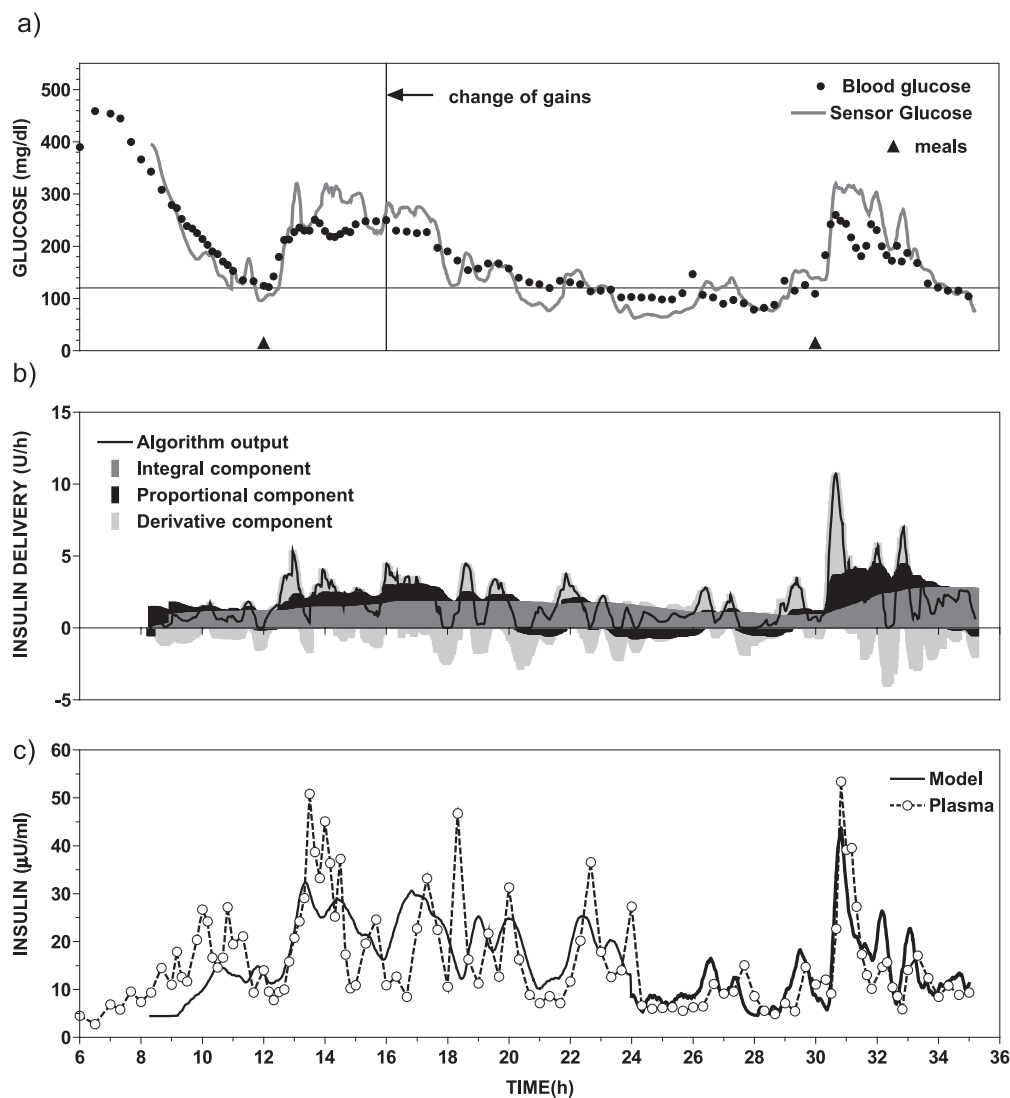


Fig. 10. External closed-loop study (SC sensor and external insulin pump, Medtronic MiniMed) in diabetic dog. (a) Plasma (solid circles) and sensor (line) glucose following initiation of closed-loop control (~ 8:30). Solid line at 120 mg/dl indicates setpoint. (b) Proportional (dark shading), integral (grey) and derivative (light grey) insulin delivery during the experiment. (c) Plasma insulin (circles) and three-compartment model prediction of insulin (solid line). Insulin delivery (pump) was suspended during the night before to generate a hyperglycemic initial condition.

glucose-tolerant humans [97], assuming a multiexponential SC insulin response curve (Fig. 8). The ratios $K_D/K_P=66$ min and $K_P/K_I=150$ min were obtained without feedback of any model-predicted plasma insulin profile ($\gamma=0$ in Eq. (3)), and the proportional gain (K_P) was calculated to deliver an amount of insulin equal to the dog's daily insulin requirement (DIR). During the experiment, the relative contribution of the PID components was separately recorded (Fig. 10b) and, following the experiment plasma

insulin (Fig. 10c), was assayed for evaluation of the “model prediction” component of the algorithm ($\hat{I}_p$ in Eq. (3)).

Three important observations can be made from the results. First, the “integral component” of the algorithm successfully established euglycemia (Fig. 10a; setpoint=120 mg/dl). This results from a steady “winding” of the integrator (depicted in medium gray; Fig. 10b); once setpoint is achieved, both the proportional (light gray) and derivative (dark gray)

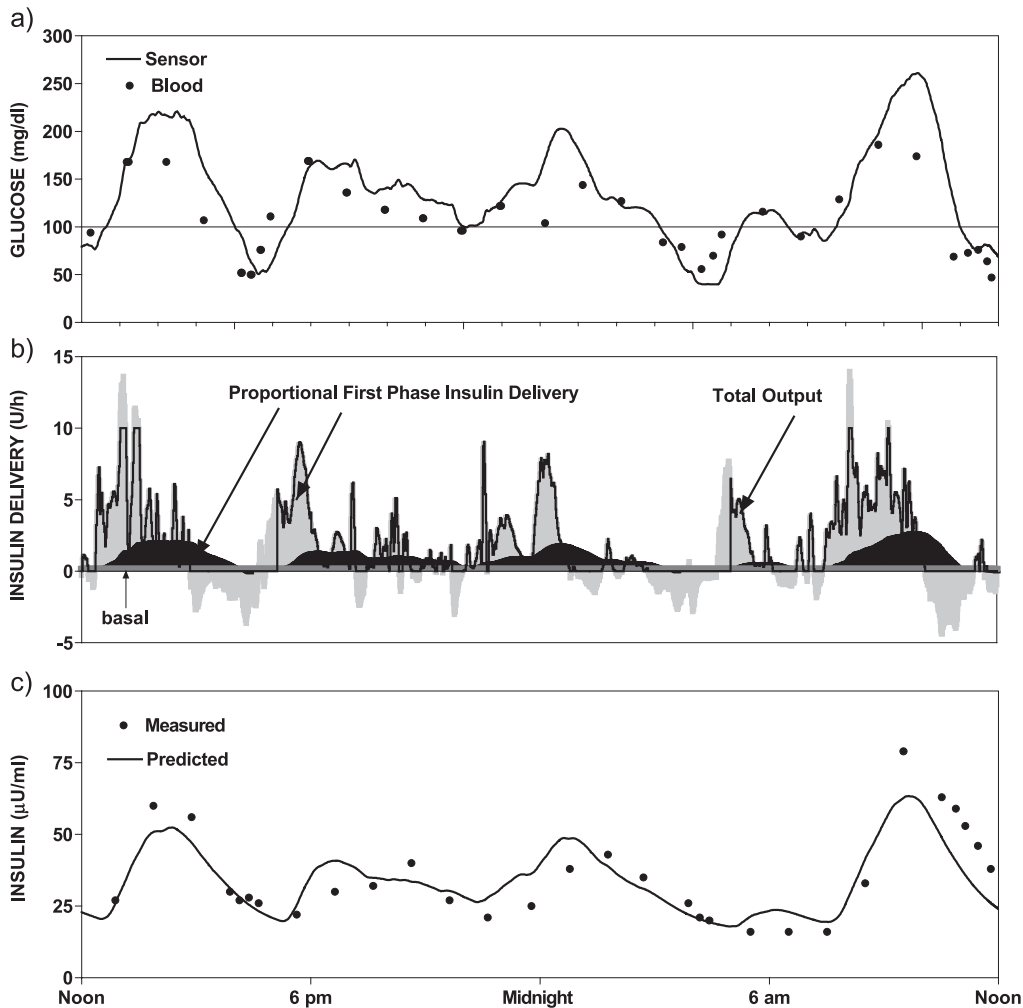


Fig. 11. Implanted closed-loop study (IV sensor and implanted insulin pump, Medtronic MiniMed) in a human subject [98,99]. (a) Plasma (solid circles) and sensor (line) glucose following initiation of closed-loop control (noon). Solid line at 100 mg/dl indicates setpoint. (b) Proportional (medium shading), basal (light shading), and derivative (dark shading) insulin delivery during the closed-loop (solid line indicates total which cannot go below zero). (c) Plasma (circles) and predicted insulin (solid line) concentrations. Study performed by Medical Research Group.

are zero. Second, the gain K_P was deemed too low; subsequent modifications of the relationship between DIR and K_P have been implemented [96], but the ratios K_P/K_D and K_I/K_P have remained unchanged. Third, the model prediction of plasma insulin concentration was not particularly good during the first day, although it improved during the second day. This result emphasizes the need to validate models with experimental data prior to incorporating them into an algorithm—particularly in situations where the predicted response may not be highly reproducible [82].

5.2. Intravenous glucose sensing with intraperitoneal insulin delivery

In humans, a fully implanted system with IP insulin delivery and IV glucose sensing has been investigated under proportional derivative control superimposed with a fixed basal rate (Fig. 11; cf. Refs. [98,99]). The sensor signal was filtered with a second-order IIR filter (cutoff frequency of 1.8 cycles/h) and the derivative estimated using a low-order FIR filter. The ratio of K_D/K_P was set at 100 min and the contribution of basal, proportional, and derivative separately recorded (Fig. 11b). Control was initiated in the morning, and the subject was given breakfast, lunch, and dinner, together with additional carbohydrate when deemed appropriate by the supervising physician. The predicted plasma insulin profile (Fig. 11c) was not used as part of the algorithm but, again, retrospectively calculated for purposes of evaluating the compensation strategy of Eq. (3).

Several observations can be made from these results relative to those of Fig. 10. First, the predicted insulin profile fit the plasma insulin more accurately (higher r^2). Second, the derivative component (first phase) was faster and contributed to a larger extent to the total delivery. The higher portion of derivative (first phase) is due to three factors: a faster absorption of insulin from the IP site versus the SC site, a higher ratio of K_D/K_P (100 vs. 66.7 min), and a faster derivative filter. The faster filter comes at the expense of a lower rejection of noise. For this system, the faster insulin absorption (cf. Fig. 8) is offset by the slower sensor response (Fig. 7). Adjusting the controller tuning to optimally emulate the β -cell in light of these differences has not yet been completed.

6. Conclusions

Individuals with diabetes have long waited for a closed-loop insulin delivery system eliminating the need to monitor their blood glucose and calculate their insulin doses. The present review attempted to cover relevant physiology and engineering research conducted in this area. Many basic physiology questions—role of first- versus second-phase insulin, contribution of portal versus peripheral insulin, etc.—have existed for several decades without adequate answers. The same can be said for many questions in sensor research, in particular those related to delays. Many approaches to automating insulin delivery have been proposed. Although there is now significant literature on each of these topics, definitive answers still elude us—there is no automated closed-loop insulin delivery system available today.

Probably, the factor most limiting in the quest for a closed-loop system has been the availability of a continuous glucose sensor. Strides have recently been made with several sensors now on the market (Medtronic; Cygnus, Menarini), or near approval (Roche/Disetronic, Therasense, DexCom, Pendragon Medical, SpectRx, etc.). As sensor technology advances, the reality of an ambulatory automated insulin delivery system approaches. The availability of these sensors will allow many control strategies, which have previously only been validated using computer models, to be investigated in vivo. Through this process, it can be expected that many questions regarding the insulin action, glucose appearance, glucose turnover, insulin secretion, counter-regulation, etc. may need to be revisited. The sheer magnitude of the number of questions that need to be addressed will hopefully promote stronger collaborative efforts between industry, academia, and funding agencies.

The stakes are high; diabetes is among the most expensive illnesses in the American health care system, with total direct and indirect costs over \$100 billion [100]. Not included in this number is the cost in human terms. Secondary complications include diabetic macro- and microvascular disease leading to coronary heart disease, nephropathy, retinopathy, and neuropathy [101–103], all of which have been shown to be reduced with improved glycemic control [104–110].

In the present review we have maintained, virtually without justification, that an effective closed-loop

insulin delivery should emulate the β -cell. Justification comes, of course, from the belief that nature has not erred in developing its glucose regulatory system. While it is unlikely that any algorithm can completely reproduce nature, we believe that emulating nature's control system is a worthy goal.

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